

Introduction

Oryza sativa, more commonly known as rice, is a global crop that is a staple in a variety of diets. As it is able to be produced at a large scale, it is generally a feasible food option for many people due to lower prices. Additionally, rice provides an excellent source of carbohydrates, which are essential for energy. The origin and cultivation of rice has a deep history, beginning in Asia. In the present day, rice cultivation has spread around the world and the crop is intertwined with a variety of cultures. However, there continue to be environmental concerns regarding rice production, considering the resources and energy required. New strategies are being utilized to continue developing this essential crop while focusing on sustainability. Rice crops also face many dangers from pests and diseases, so management practices are necessary in order to not only maintain crop yields but also protect the communities that heavily rely on rice.

History and Origin of Rice

There are some debates over the origins of rice cultivation. The general consensus is that evidence points towards rice cultivation beginning in Asia, but the specific timeframe and area are still not completely clear. In *Archaeological and genetic insights into the origins of domesticated rice*, Briana L. Gross and Zhujun Zhao state there is evidence of rice cultivation in India dating to 5000 BP, but archaeological sediments in the Yangtze River Valley point to origins of rice domestication earlier in China (Gross and Zhao, 2014, pg 5). Megan Sweeney and Susan McCouch also support this in *The Complex History of the Domestication of Rice*, explaining how a silicon microfossil from rice called phytoliths, dating back to 11,000 BC, and various remains of rice have been found in multiple sites of the Yangtze River Valley (Sweeney and McCouch, 2007, pg 2). In *A Journey to the West: The Ancient Dispersal of Rice Out of East*

Asia, authors claim that rice has two different phenotypic lineages, *O. sativa* ssp. *japonica* and *O. sativa* ssp. *Indica*. It is likely rice was primarily cultivated in East Asia, and then spread through mountain passes to South Asia. Then, the lineages were separately cultivated in eastern China and northern India, respectively (Spengler III et al., 2021, pg 2). This explains that rice cultivation may have begun separately in East and South Asia, considering the distinct lineages. Using linguistic data, historical source, and evidence of rice remains, Sweeney and McCouch traced the expansion of rice cultivation. Through trade networks and migration, rice spread to Western Asia and eventually to Africa and Europe (Sweeney and McCouch, 2007, pg 4). In the United States, rice cultivation began in 1646 in Virginia (Heinrichs). In the present day, rice is a global staple food important to make cultures and diets.

Sustainability Considerations

Sustainability can broadly be defined as the ability to fulfill present day needs without negatively affecting the ability to fulfill future needs. To understand the sustainability of rice, it is important to primarily understand how rice is grown. The American Farm Bureau Foundation for Agriculture explains how rice is traditionally grown by flooding fields. This has benefits of reducing soil erosion and the appearance of weeds. After the plant grows, the fields are drained so the plants can be harvested (American Farm Bureau Foundation for Agriculture). This practice depicts that there is a large need for energy and natural resources such as water for rice cultivation. In *Exploring the Impact of Alternate Wetting and Drying and the System of Rice Intensification on Greenhouse Gas Emissions: A Review of Rice Cultivation Practices*, James Dahlgren and Adam Parr explore the sustainability of rice cultivation and the benefits of sustainable techniques. Primarily, they discuss the large greenhouse gas emissions that result from rice cultivation, especially by using traditional practices such as flooding rice fields.

Flooding causes a process called methanogenesis, in which methane is produced as a by-product of the metabolism of archaea in anaerobic soil (Dahlgren and Parr, 2024, pg 1). The greenhouse gas emissions show that rice is not a sustainable crop. Greenhouse gases contribute to global warming, which results in harmful impacts to the environment. This results in negative impacts to the ability to fulfill future needs. However, measures can be taken to develop more sustainable production.

In *Rice cultivation and processing: Highlights from a life cycle thinking perspective*, authors further evaluated the sustainability of rice by reviewing 76 Life Cycle Assessment studies. They found that methane emissions from rice paddies was the primary negative environmental impact of rice production (Ahmad et al., 2023, pg 7). This further supports that the cultivation of rice causes large emissions, adding to the carbon footprint of agriculture. The UK Research and Innovation organization also points to traditional rice milling, which uses stones to strip rice. This generates more heat and results in 30 to 50 percent of the crops being wasted (UK Research and Innovation, 2025). Not only does rice farming have significant environmental impacts, the later stages before rice can be sold also leave a large carbon footprint. As rice is a widely cultivated crop, it is essential to consider the significance impacts its cultivation has on the environment.

Sustainable Practices

Some strategies have been developed in order to make rice production more sustainable. Dahlgren and Parr discuss Alternate Wetting and Drying (AWD), which is a process that involves waiting between one to ten days before flooding a field. This practice can primarily help reduce the water needed in rice cultivation and also stop the process of methanogenesis (Dahlgren and Parr, 2024, page 3). AWD proves to make rice production more sustainable by

saving water and lowering gas emissions. The authors also discuss the System of Rice Intensification (SRI), which is planting seedlings in a grid pattern to reduce competition (Dahlgreen and Parr, 2024, page 3). Reduced competition can lower the need for fertilizers, which can protect the soil and biodiversity. The UK Research and Innovation organization shares a new rice milling machine, called Koolmil, that is more environmentally friendly than older, conventional mills. Due to gently processing the rice grains as a fluid, the Koolmill is able to use 80% less energy than traditional milling technology (UK Research and Innovation). Evolving technologies and strategies can help ensure rice becomes a more sustainable crop.

Grown

Rice is grown in many areas globally. In *Management of Rice Insect Pests*, Heinrichs describes there are two major ways it can be produced, which includes a dry system or a wet system. The dry system involves rice crops being grown on dry soil, while the wet system involves flooding the field (Heinrichs, 2015, pg 1). The United States Department of Agriculture describes the requirements needed to grow rice, including smooth land, a large water supply, and hot, humid days with cooler nights (U.S. Department of Agriculture). This results in areas with tropical climates being ideal to grow rice.

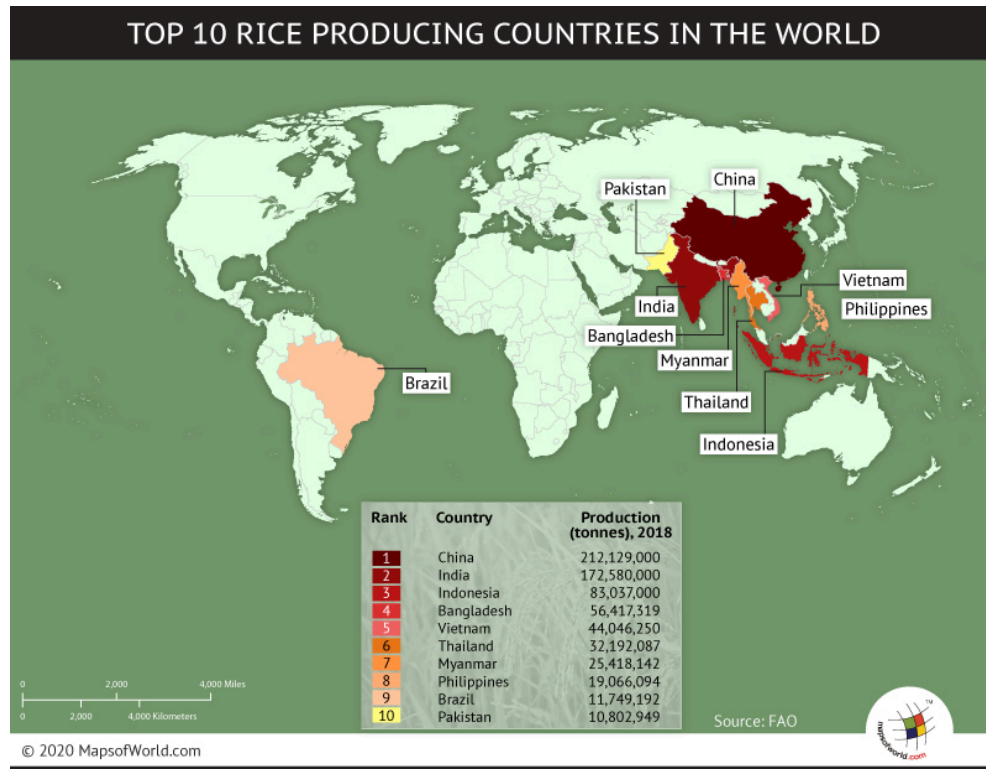


Fig 1. Map depicting leading countries in rice production in 2018, measured by tonnes produced. The majority is made up of Asian countries, with China, India, and Indonesia in the top 3 (Maps of the World, 2020).

This is supported by Figure 1, which depicts China and India to be the top producers of rice globally, producing over 212 million tonnes and 172 million tonnes, respectively in 2018. This is a result of these regions having a warm and humid climate, which is the optimal environment for rice cultivation. Heinrichs also describes rice to be adaptable to a variety of environments, but largely being grown between 53° N to 40° S latitude (Heinrichs, 2015, pg 2). This depicts that although rice may be a crop that can adapt to different climates, it generally best flourishes in areas that have moist soil and a humid climate that can support the significant amount of moisture the crop requires.

Pests and Diseases

There are a variety of pests and diseases that pose a threat to rice crops. In *Sustainable Insect Pest Management Options for Rice Production in Sub-Saharan Africa*, authors how the

pests can vary based on the stage of crop production and management techniques. They focus mainly on rice pests in Africa, where the African rice gall midge is an incredibly damaging pest. This pest constricts the growth of rice plants by creating gall formations. Other pests are the white and pink borers, which cause deadhearts by piercing into the stems, which causes losses in crop yields (Pegalepo et al). Another type of pests are defoliators, which feed on the foliage of plants and cause them to have reductions in photosynthetic capacity. Just as rice flourishes in warm and humid climates, these pests also rapidly multiply in this type of climate. In

Management of Rice Insect Pests, Heinrichs describes more rice insects. The rice water weevil is a type of termite that eats the lives of rice plants. They do not cause much damage, but the larvae diminish root systems by feeding on roots. This causes the plant to not grow properly. Heinrichs also describes leadhoppers and plant hoppers, which attack both the base and aerial part of the plant. They attack the plant by sucking the sap out of the plant's tissues, causing the plant to dry (Heinrichs, 2015, pg 3). These pests are a large concern as not only do they lower crop yields, but a disruption in the rice supply can affect economies. Heinrichs states that in 1733, there was a planthopper outbreak in Japan, which caused 12,000 to die from hunger due to damages to rice plants. (Heinrichs, 2015, pg 5). This situation depicts how people are incredibly reliant on rice and how much decreases in crop yields can affect the economy and human health.

There are a variety of diseases that can also affect rice crops. Some bacterial diseases include *Xanthomonas oryzae*, or bacterial leaf streak, and *Xanthomonas oryzae* pv. *Oryzae*, also known as rice bacterial blight. Bacterial leaf streak has symptoms of tears in the veins of leaves, which can enlarge and cause the leaves to die. It can spread through seeds and water. Rice bacterial blight is similar, in which it creates lesions on leaves and dry the leaf out, causing the plant to die (Plant Village, pg 2). Some fungal diseases include *Fusarium moniliforme*, or

Bakanae, and Brown spot, or *Cochliobolus miyabeanus*. Bakanae is a disease that is spread through seeds. It causes the seeds to grow abnormally, causing the plant to grow incorrectly. Brown spots are seen by oval spots on the plants, and can cause abnormally colored rice grains. These diseases can quickly spread through rice crops and ruin crop yields (Plant Village, pg 3). If pests and diseases become an unmanageable problem, it can significantly affect the crop supply and can have adverse effects on people and the economy.

Pest Management

In *Management of Rice Insect Pests*, Heinrichs describes how previous, traditional practices would focus on the use of insecticides. A newer approach, called integrated pest management, combines a variety of methods to battle against rice pests. This has been a very effective technique in reducing the indirect effects of pesticides (Heinrichs, 2015, pg 7). Pegalepo expands on this by providing an example of the Integrated Rice-Fish System, which combines fish farming and rice cultivation to decrease pests. IPM also refers to cultural practices that have been tried and tested for many years (Pegalepo, 2025, pg 5). One example is draining the field to reduce the aquatic casework larva. Additionally, practices such as weeding and land preparation can help with reducing the number of pests (Heinrichs, 2015, pg 5). There are a variety of indigenous farming practices that can be utilized to protect plants against pests.

Insecticides can also be utilized as a measure, but they are often very expensive to use. There are also a variety of concerns with the adverse effects that insecticides have on the environment, including water contamination and soil degradation (Heinrichs, 2015, pg 7). When considering the sustainability of rice, it is also important to consider the adverse impacts of rice production also impact the environment. There are also a variety of pathogens that can help reduce rice pest populations, such as *Metarhizium anisopliae*. This pathogen helps decrease

populations of planthoppers and leafhoppers. Pegalepo and contributors also explore a variety of pest management options, including crop rotations. They state that in African countries, rotating rice with crops such as cowpeas or maize helped reduce populations of rice pests. (Pegalepo, 2025, pg 7).

They also discuss selective breeding to breed rice plants that have resistance to pests. Using the genetic diversity of different plant varieties can help with battling against harmful pests. They also go more in depth on synthetic pesticides, which are created to kill weeds, insects, and pathogens. Synthetic pesticides have proven to be useful, saving over 50% of global agriculture. Biological controls are also an option for pest management. Introducing natural predators, such as *Trichogramma* wasps in Nigeria, can be a more environmentally friendly alternative to insecticides (Pegalepo, 2025, pg 9).

Price

The Federal Reserve Bank of Saint Louis depicts the average price of rice in U.S. cities, depicting fluctuations over the years, and a general increase since 2020. Figure 2 shows the current market price as of 2025 is around 1.07 dollars per pound (Federal Reserve Bank of Saint Louis, 2025). The UN Environment Programme explains that rice generally has low prices due to being mass produced and their being a large global supply (UN Environment Programme, 2019). As supply of a product increases, the market price decreases. However, there are still certain types of rice that are more expensively sold than others, such as Jasmine rice. This is because certain types of rice require more tedious controls or having special flavors or aromas. In *Stock-to-Use Ratio (STU) and Price of Rice: Deciphering the Relationships*, authors explore the stock-to-use ratio of rice, which is a ratio of stocks to the total use of rice. They combined data from the top 16 rice producing counties, and found there to be an inverse relationship between

STU and rice prices. This means when STU is high, export prices of rice are generally lower (Maloney, 1991, pg 6). This is related to the idea of scarcity causing increases in rice prices, while abundance causes decreases. It also points to how trade policies, such as export bans, may affect STU and in turn rice prices. It depicts the relationship between the government and accessibility to crops.

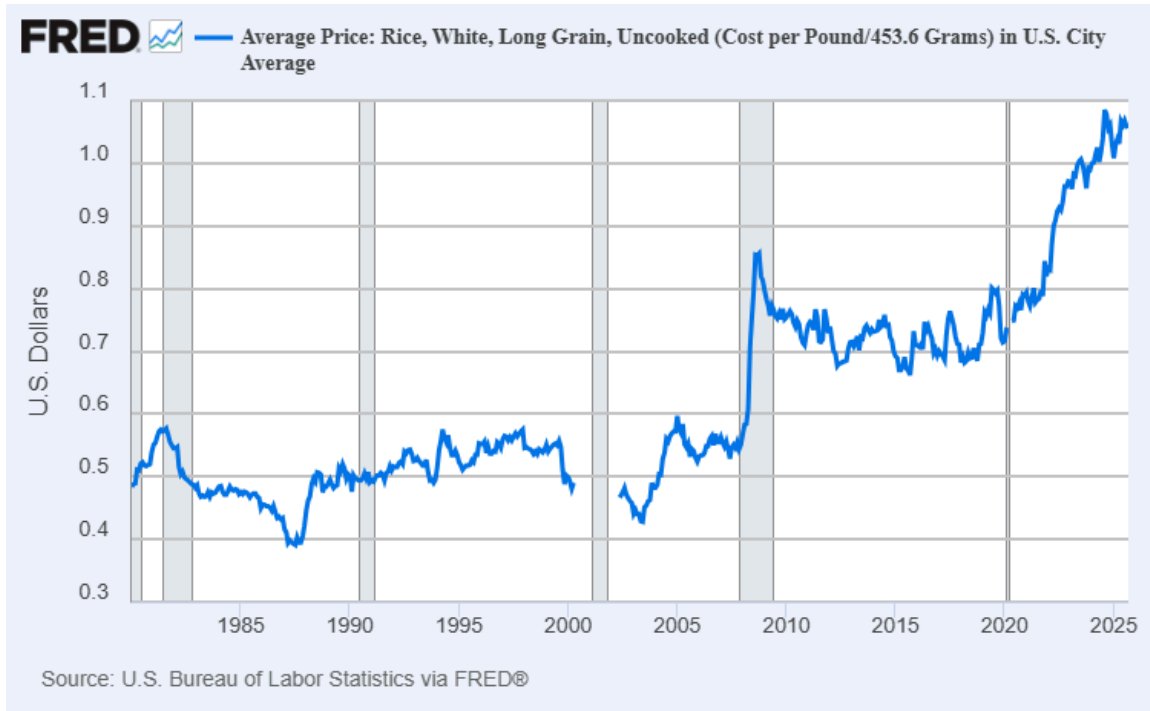


Fig 2. Graph showing average price of white, long grain, uncooked rice in United States cities. It shows the price in U.S. dollars per pound in relation to years.

Conclusion

As discussed, rice cultivation began in Asia, particularly China, and has spread globally. The large supply of rice, especially from leading exporting countries such as China and India, has caused the crop to have relatively low marketing prices. However, the large-scale cultivation also results in there being a variety of harmful environmental impacts, including a large use of water as well as increased greenhouse gas emissions. Sustainable practices, such as new milling techniques, System of Rice Intensification, and Alternate Wetting and Drying are used in order to

make rice production more environmentally friendly. Since rice is such a staple crop, it is important to find management practices that will protect rice plants from a large variety of insects and diseases that can affect them. The cultivation and global use of rice today depicts the significant power of crops to tie people and nations together.

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